

Towards Physics-guided Generative Foundation Models

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Abstract

A generative model is a statistical model of the joint probability distribution on a given observable variable and target variable, which can be used to "generate" random instances (outcomes) of an observation. When trained on broad and diverse data, it serves as a foundation model that reduces the training resources (e.g., time, energy, labeled samples) required for a wide range of downstream tasks. Due to recent success, such as the generative pre-trained transformer (GPT), there is ongoing excitement about foundation models. However, these models often struggle with out-of-distribution generalization and may yield unrealistic or physically invalid outputs. To address these limitations, we envision physics-guided generative foundation models (PgGenFMs), which integrate broad physical knowledge into foundation models to improve reliability and applicability. We also present a taxonomy comparing PgGenFMs to traditional models and outline strategies for embedding physical knowledge into data, training, and model design.

Keywords

Physics-guided Generative Foundation Models (PgGenFMs), Physics-guided Task-specific Models (PgTSMs), Foundation Models (FMs)

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1 Introduction

Recent advances in AI have driven significant progress in science and technology, as highlighted by recent Nobel Prizes in chemistry and physics that recognized AI's impact, including its role in

protein structure prediction through tools like AlphaFold and developments in neural networks [28, 29]. Central to these advances is the *generative model*, which learns a parameterized probability distribution, $p_{\theta}(X)$, to approximate the true distribution of a dataset, $p_{\text{data}}(X)$. This allows it to generate new data instances by sampling, often conditioned on some context Y by modeling $p_{\theta}(X|Y)$. When trained on broad and diverse data, they serve as foundation models (FMs) that reduce the training resources (e.g., time, energy, labeled samples) required for a wide range of downstream tasks. Notable examples include GPT models, BERT, DALL-E, Stable Diffusion, and CLIP. These foundation models, built with billions of parameters, provide powerful tools for tasks such as dialogue, translation, code generation, and image interpretation, enabling researchers to address increasingly data-rich and complex problems.

The success of FMs has been driven by the availability of large-scale datasets, increased computational resources, and breakthroughs in deep learning architectures. Techniques such as transformer architecture [42] and contrastive representation learning [7] have significantly contributed to this progress. For a complete overview, we refer the reader to [47].

Despite their versatility, purely data-driven foundation models (FMs) exhibit significant limitations in scientific and engineering domains. These models often struggle with out-of-distribution scenarios because they rely solely on statistical patterns present in the training data. In addition, they fail to respect fundamental physical principles and laws of nature, which can result in physically implausible predictions. Furthermore, data-driven FMs typically function as black boxes, offering limited interpretability and no guarantees that their learned representations align with physical or causal relationships. Recent efforts have introduced preliminary designs for physics-guided foundation models (PgFMs), including the Aurora model for the Earth system [23] and work by Yu et al. [46] in aquatic science, where the models primarily relied on conventional architectures such as encoder-decoder frameworks and long short-term memory networks.

To address these limitations, we envision physics-guided generative foundation models (PgGenFMs) as a more comprehensive class of generative models that incorporate physical principles, encompassing both broad and narrow domain knowledge, as well as fundamental scientific laws, across a variety of model architectures. These include variational autoencoders (VAEs), generative adversarial networks (GANs), normalizing flows, diffusion models, energy-based models, and autoregressive models [1, 3, 4].

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Contributions. Our primary contributions include the following: (1) This paper formally defines the concept of physics-guided generative foundation models (PgGenFMs) and lists a few methods to incorporate broad-domain physical knowledge. (2) This paper provides a taxonomy that compares PgGenFMs with other machine learning methods as well as a concrete example that illustrates the limitations of existing FMs and the need for PgGenFMs.

Background: Figure 1 outlines the lifecycle of foundation models, from large-scale pre-training to inference. The process begins by training a large-scale model (e.g., with billions of parameters) on billions of images and trillions of tokens. This stage produces a general-purpose foundation model with broad generalization capabilities. The model can later undergo fine-tuning using domain-specific data and human-annotated examples to be customized for downstream tasks, such as educational content generation.

At inference time, the specialized model receives a query and produces a response with low latency and cost. Foundation models also support zero-shot and few-shot settings, where task adaptation does not require parameter updates. In zero-shot settings [43], the model performs tasks using only natural language instructions without labeled examples. For example, it can translate text, summarize documents, or classify sentiment based solely on the prompt structure. Few-shot learning [6] improves performance by providing a small number of demonstrations within the prompt, such as two or three pairs of questions and answers, before posing a new question. These abilities stem from large-scale pre-training, eliminating the need for task-specific parameter updates. Retrieval-augmented generation (RAG) further enhances performance by retrieving relevant documents or passages at inference time. The model integrates this external information into its response, improving the accuracy of the facts and supporting knowledge-intensive tasks in dynamic or specialized domains [24].

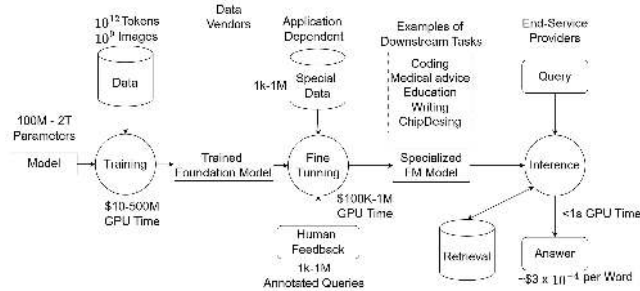


Figure 1: Traditional foundation model pipeline [9].

2 Challenges and Limitations of Data-Driven Foundation Models

Challenges: The first key challenge is their struggle with out-of-distribution situations [45]. This occurs because they solely rely on statistical correlation in data, and their benefits to downstream applications may thus be limited. This shortcoming is particularly problematic in specialized tasks that require nuanced understanding and adaptation to specific domain knowledge, such as climate science and healthcare. For instance, the geospatial foundation model (Prithvi) [17], trained exclusively on satellite imagery data from Landsat and Sentinel with multi-spectral bands, may fail to

generalize to other data types, such as hyperspectral imagery or newer satellites with different spectral resolutions.

Second, these models may fail to adhere to fundamental physical principles, such as energy conservation and motion dynamics. In the absence of physics-based constraints, these models can produce outputs that are unrealistic or physically infeasible. For example, models trained to estimate energy consumption and generate velocity profiles using onboard vehicle diagnostics and trajectory big data display rapid and unrealistic variations in velocity. Figure 2(a) shows velocity profiles without a jerk penalty that exhibit sharp peaks and drops, which reflect abrupt speed changes not typical in realistic driving scenarios. Figure 2(b) demonstrates that omitting a jerk penalty results in excessive jerk values far beyond passenger comfort thresholds [11].

Third, their black-box nature reduces transparency and makes predictions difficult to interpret [44]. These challenges underscore the need for physics-guided generative foundation models (PgGenFMs), which integrate scientific laws and physical constraints to improve prediction reliability, robustness, and domain-level trust.

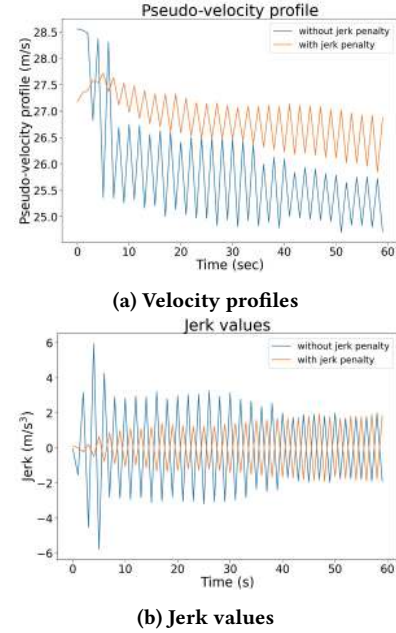


Figure 2: Neural network-generated velocity profiles and corresponding jerk values.

Limitations: One significant limitation involves the *computational cost* incurred in the training of large-scale models. These models demand immense energy resources and substantial financial expenditure, which often makes them infeasible outside of resource-rich settings [30, 41]. A second issue pertains to *sample complexity*. These models typically require vast amounts of training data, which limits their suitability in domains where data acquisition is prohibitively expensive, experimentally constrained, or inherently sparse. Such conditions are common in fields such as biomedicine, climate science, and materials engineering [20, 27]. Most importantly, purely data-driven models often operate as “*black boxes*”, with limited interpretability and no guarantees that their

learned representations reflect fundamental physical or causal principles. This opacity introduces substantial risk, particularly in high-stakes applications where consistency with physical laws remains essential for safety, reliability, and trust [20, 33]. These limitations collectively highlight the *imperative to incorporate domain knowledge* which must be integrated into model design to ensure accurate, interpretable outputs consistent with real-world laws.

Physics-Guided Task-Specific Models (PgTSMs) have shown strong performance in scientific and engineering domains by embedding domain-specific physical laws into model architectures [31, 44]. Designed for narrowly defined problems, these models often surpass purely data-driven approaches in accuracy, sample efficiency, and adherence to physical principles. Applications in fluid dynamics, climate modeling, and structural mechanics demonstrate that incorporating known constraints improves generalization in data-scarce settings and ensures consistency with conservation laws [20, 31]. However, the limited generalizability of these models remains a constraint, as each new task typically necessitates an independent, resource-intensive development cycle involving manual feature engineering, expert-driven constraint formulation, repeated hyperparameter tuning, and extensive domain-specific data collection.

PgGenFMs offer a more generalizable framework than PgTSMs models by embedding broad domain knowledge and universal physical laws, such as conservation of mass, energy, and momentum, into the pre-training phase of the foundation model. This design reduces sample complexity, improves interpretability, and promotes physical consistency across a wide range of downstream tasks [20].

3 Vision

We envision a physics-guided generative foundation model (PgGenFM) defined as a foundation model designed to incorporate broad domain knowledge and a general understanding of the fundamental concepts and principles of the domain. For example, popular knowledge graphs and taxonomies [18] can strengthen foundation models for a wide range of downstream applications. Unlike the traditional FMs that learn implicitly from data, a PgGenFM is explicitly structured to respect foundational scientific laws. Conservation laws such as those governing mass and energy apply to many physical science tasks. The concepts of speed, acceleration, and Newton’s laws are applicable to problems involving moving objects. Maxwell’s equations are essential in tasks related to electromagnetism. A generalizable model of bio-geochemical processing (e.g., carbon and nitrogen cycles, nutrient transport, soil–atmosphere interactions) may be used as a PgGenFM with extensive training.

Figure 3 presents the proposed PgGenFM framework. It refines the standard data-driven foundation model pipeline by incorporating physical knowledge throughout model development. This includes both *broad-domain knowledge*, which reflects general scientific principles such as energy conservation and Maxwell’s equations, and *narrow-domain knowledge*, which involves constraints specific to a particular application. Broad-domain knowledge shapes data representations and model design, ensuring alignment with known physical laws. Narrow-domain knowledge addresses localized or task-specific dynamics. Unlike most traditional models that are primarily built upon self-supervised learning and encoder-decoder architectures, PgGenFMs consider a broader family of generative models (e.g., normalizing flows, diffusion models [3, 16]) for

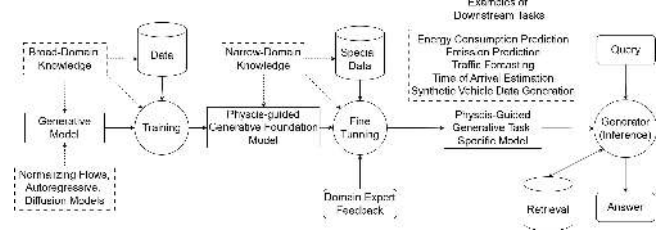


Figure 3: Proposed system architecture for the Physics-guided Generative Foundation Model.

embedding physical knowledge. The resulting PgGenFM captures a physically meaningful structure, which improves generalization and interpretability compared to purely data-driven approaches.

To support downstream tasks such as traffic emission prediction, energy consumption prediction, or traffic forecasting, the PgGenFM undergoes fine-tuning using smaller, specialized datasets. During this phase, *narrow-domain* knowledge from expert input, simulation models, or high-resolution measurements informs model adaptation. Domain-expert feedback further refines the model by introducing labeled examples or custom-made corrections. For example, aerodynamic drag predictions may use wind tunnel data, while vehicle energy consumption estimation can rely on logs reflecting Peukert’s law [12]. This law, which describes the non-linear discharge behavior of batteries under varying current loads, applies to specific types and conditions of powertrains. Rather than embedding such fine-grained laws during pre-training, broader energy dynamics, such as the general relationship between current and energy use, can guide the initial model. Specific principles like Peukert’s law then become relevant during fine-tuning for tasks such as estimating electric vehicle range under operational constraints. This process yields models with both high predictive accuracy and strong physical consistency and generalizability.

Broad vs. Narrow Datasets for PgGenFMs: The distinction between broad and narrow physics knowledge also appears in the datasets used to train PgGenFMs. Pre-training typically involves large-scale, diverse datasets that capture general physical principles. Examples include highway traffic flow datasets, vehicle trajectory datasets such as NGSIM [8], and energy datasets like FASTSim simulations [5], calibrated to US Environmental Protection Agency standards. These datasets help the model internalize broad physical relationships, such as energy conservation and speed–density dynamics. For downstream tasks, more specialized datasets are required. For instance, aerodynamic drag models may be refined using vehicle-specific wind tunnel data, while battery discharge predictions may be fine-tuned with high-resolution battery management system logs that capture real-world Peukert’s law behavior.

3.1 How Can Domain Knowledge Be Embedded in Generative Models?

A variety of approaches can be used to integrate physical knowledge into generative models. *Physics-constrained learning* can enforce real-world constraints by introducing domain rules into the learning process through the loss function or regularization terms that safeguard the physical feasibility of model predictions (e.g., [25]). Similarly, *architecture-level integration* (e.g., [19, 25]) incorporates domain principles directly into model structures, aligning network

design with physics-based rules and patterns. As noted [25], we recently demonstrated that incorporating a physical constraint, such as a jerk penalty, into velocity profiling reduced the sharp fluctuations common in unconstrained models, as shown in Figure 2. This adjustment yielded smoother outputs that more accurately reflect realistic physical behaviors and practical driving dynamics. Another strategy involves using *physics-based surrogate augmentation* during pretraining, where simulation-driven or surrogate model outputs are used to enrich the training set with physically consistent examples. Finally, *hybrid architectures* combine data-driven components with physics-informed modules, allowing models to benefit from both empirical flexibility and physical constraints.

Comparison with Related Models: Table 1 compares PgGenFM with related models based on the criteria of breadth of training data, the extent of domain knowledge, and the range of downstream tasks. In traditional task-specific (i.e., machine learning) models are trained on narrow datasets, which limits them to a restricted range of downstream tasks and generally lacks deep integration with specific scientific or domain knowledge. Foundation models (FMs) are trained on broad datasets, enabling them to support a wide range of downstream tasks; however, they are typically data-driven and have limited or non-existent integration of broad domain knowledge. Physics-guided task-specific models (PgTSMs) integrate deep but narrowly focused domain knowledge (e.g., scientific) into the learning process [10, 21]. They concentrate on specific tasks and datasets, leading to the development of specialized models.

Table 1: Comparison of PgGenFMs with Related Models

Levels	Technique	Training Data	Scientific Knowledge	Downstream Tasks Range
1	task-specific models	–	–	Narrow
2	FMs	Broad	–	Wide
3	PgTSMs	–	Narrow	Narrow
4	PgGenFMs	Broad	Wide	Wide

3.2 A Few Open Questions in PgGenFMs

In the context of physics-guided architecture design, an open problem is determining where domain knowledge should be incorporated into the architecture. Consider the encoder-decoder framework, commonly used in many foundation models. Physical knowledge may be introduced through different components, such as shaping the encoder to guide representation learning or designing the decoder to improve the interpretability of model outputs. Which stage of the architecture benefits from domain knowledge? Should physical constraints influence the internal representation, or are they more effective when applied at the output stage?

Other open issues include how to define optimization and choose algorithms for the training stage of PgGenFMs. Traditional algorithms such as backpropagation often fail to satisfy hard constraints such as mass balance, which may require the use of alternative strategies. A clear distinction is also needed between broad domain knowledge and task-specific information. How can physical constraints be enforced during training? When should models rely on general principles rather than task-specific rules?

Mitigating biases requires that training data cover a wide range of scenarios, including occlusions (e.g., cloud cover), rare and extreme events (e.g., polar regions), subpixel objects, and slow geological processes. Data quality issues also arise from gaps, anomalies, hotspots, and deceptive behaviors where objects broadcast false

location signals to mislead end-users. What methods can ensure equitable representation across diverse conditions? How can models detect and adapt to intentionally misleading or incomplete data?

Another challenge in ensuring that PgGenFMs can account for variations in data and feature characteristics in different scales, spaces, and locations. In geospatial contexts, the objective is to capture general spatial trends while preserving location-specific information. What types of domain knowledge should be integrated into PgGenFMs? To what extent does this integration influence the model’s ability to adapt to location variability?

PgGenFMs may require adaptation to account for location-specific factors, such as regional conditions or contextual variability. This motivates the development of localized PgGenFMs that incorporate geographic information tailored to specific downstream tasks. However, this raises open questions: Do location-specific models introduce biases that affect geospatial applications? Does memorization of localized patterns limit the generalization capacity of global models? As datasets continue to grow, these concerns remain critical for maintaining robustness and mitigating bias [26].

4 Conclusion and Future Work

PgGenFM promises to enhance foundation models by systematically integrating domain-specific physical knowledge, improving performance, robustness, and interpretability for diverse applications. Future research holds promise for promising advancements in AI across various fields, including healthcare [15], geospatial analysis, and engineering. We plan to develop and evaluate PgGenFMs, integrating domain knowledge such as sample complexity.

Advancing PgGenFM in future work will involve developing hybrid frameworks that combine the adaptability of retrieval-augmented generation [24] with real-time updates to optimize domain adaptation and reduce training resource demands. We will also explore PgGenFMs in the context of recent multimodal foundation models [32], focusing on the integration of physical knowledge [22, 34–40] from various input sources and its impact on training procedures and downstream applications. We will also explore additional criteria for comparing PgGenFMs with other machine learning models, including training methods and challenges such as model complexity, computational requirements, location dependency, and calibration needs [14, 15]. In the long term, we aim to address challenges of explainability [2, 13], robustness, bias, data quality, and adaptability through techniques that trace how domain knowledge shapes predictions, thereby enhancing model transparency. Beyond these directions, future work must advance from conceptual framing to concrete algorithms and architectures, validated on diverse datasets, while addressing implementation challenges (e.g., embedding physics, managing constraints, ensuring scalability) and establishing the viability and transparency of PgGenFMs.

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